technological innovations, and many scientists spent most of their time working on arms, medicines and machines for their imperial masters. A common saying among European soldiers facing African enemies was, ‘Come what may, we have machine guns, and they don’t.’ Civilian technologies were no less important. Canned food fed soldiers, railroads and steamships transported soldiers and their provisions, while a new arsenal of medicines cured soldiers, sailors and locomotive engineers. These logistical advances played a more significant role in the European conquest of Africa than did the machine gun.

But that wasn’t the case before 1850. The military–industrial–scientific complex was still in its infancy; the technological fruits of the Scientific Revolution were unripe; and the technological gap between European, Asiatic and African powers was small. In 1770, James Cook certainly had far better technology than the Australian Aborigines, but so did the Chinese and the Ottomans. Why then was Australia explored and colonised by Captain James Cook and not by Captain Wan Zhengse or Captain Hussein Pasha? More importantly, if in 1770 Europeans had no significant technological advantage over Muslims, Indians and Chinese, how did they manage in the following century to open such a gap between themselves and the rest of the world?

Why did the military–industrial–scientific complex blossom in Europe rather than India? When Britain leaped forward, why were France, Germany and the United States quick to follow, whereas China lagged behind? When the gap between industrial and non-industrial nations became an obvious economic and political factor, why did Russia, Italy and Austria succeed in closing it, whereas Persia, Egypt and the Ottoman Empire failed? After all, the technology of the first industrial wave was relatively simple. Was it so hard for Chinese or Ottomans to engineer steam engines, manufacture machine guns and lay down railroads?

The world’s first commercial railroad opened for business in 1830, in Britain. By 1850, Western nations were criss-crossed by almost 25,000 miles of railroads – but in the whole of Asia, Africa and Latin America there were only 2,500 miles of tracks. In 1880, the West boasted more than 220,000 miles of railroads, whereas in the rest of the world there were but 22,000 miles of train lines (and most of these were laid by the British in India).[5] The first railroad in China opened only in 1876. It was 15 miles long and built by Europeans – the Chinese government destroyed it the